UNIT 1 PROJECT: SUSTAINABLE DESIGN

The Heat Exchange

A student learning hub on the Clifton Campus car park



The proposal placed on its site in Google Earth Pro. Image: Google Earth.

Design Outline

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Executive summary

This Design Outline presents **The Heat Exchange**, a student educational resource centre built to replace the existing Clifton Campus car park. The design responds to the Unit 1 Sustainable Design brief by integrating efficient heat-management solutions, renewable energy sources and a renewed natural landscape (Nottingham Trent University, no date). Through those choices we address clusters of the United Nations Sustainable Development Goals: Goals 3–5 (health, education and equality), Goals 6–9 (water, energy, work and infrastructure) and Goals 13–17 (climate action and partnership) (United Nations, 2015). How individual features map to specific targets is set out in Section 4.

We express the project in several accessible formats, so different audiences can engage with it:

- A **physical model** at 1:1000 of the surrounding terrain, beside a 1:100 "doll's house" building whose floors lift off so visitors can see rooms and furniture on each level.
- An **interactive web twin**, hosted on GitHub Pages and linked from a QR code on the poster, so people can walk the landscape and building in their own time.
- This **Design Outline**, plus four **laser-engraved plywood walls** that slot around the site model and carry the quick-read project facts.
- A **Google Earth** placement of the building in its exact campus position.

Those routes are deliberate. They show that a design can be communicated interactively, across media, rather than only as a paper poster.

Table 1 Key figures (from the design model unless noted)

Quantity	Value	Basis
Gross internal area	720 m^2	24 × 14.4 m main block, two storeys, plus 6 × 9.6 m cycle hub
Storeys / height	2 storeys; 3.6 m floor-to-floor; about 9.4 m to roof peaks	Structural grid
Fit-out (priced items)	£88,270 (£123/m^2)	Itemised room schedule from the 3D model
Electricity from roof PV	about 46.9 MWh/yr (49 kWp, 114 panels at 25°)	PVGIS for this location (European Commission Joint Research Centre, no date)
Hypothetical project budget	about £2.4 million	£2,800/m^2 GIA construction benchmark + fit-out + PV + plant allowance Section 5

SECTION 1

Introduction and site

1.1 The brief and design aims

The project brief asks teams to design a sustainable work or living space that addresses one or more of the United Nations Sustainable Development Goals, to integrate a variety of sustainable technologies, and to present the design at a poster-display event (Nottingham Trent University, no date). Required outputs are technical drawings, a physical poster, and this Design Outline, with research supported by Harvard referencing (Nottingham Trent University, no date).

Our response is guided by three linked aims:

1. **Reduce operational energy** through efficient fabric, lighting and controls.
2. **Generate and recover energy** on site with roof photovoltaics and waste-heat recovery from an edge server room.
3. **Serve students** in a hospitable building that offers quiet study, teaching, making, socialising and wellbeing spaces, set in an expanded green landscape.

1.2 Location

The proposal sits on College Drive, Clifton, Nottingham NG11 8NS, on the existing Students' Car Park. The building's south-west corner is at approximately 52.9135° N, 1.1839° W (British National Grid 454976 E, 335433 N). We chose the site for the large green space around it, the relatively flat ground, and the chance to expand campus ecosystems rather than cut into them. Groundworks would be limited compared with a steep or densely built plot; under the plan the car park becomes meadow, paths, a pond and outdoor amenity space, with the new building settled in that landscape.



Figure 1 The Heat Exchange placed on the Clifton Campus car park in Google Earth Pro

The proposal is intended to sit comfortably with the surrounding campus, which is already strong in landscape quality. NTU's campuses have been recognised with the Green Flag Award for the management of green space (Nottingham Trent University, 2026).

1.3 Existing condition and proposal

At present the site is a hardstanding car park, with a rough boundary of hedges, set back from the main campus approach behind a barrier. The intention is to replace most of that surface with a carefully developed green space and to place the new building within it.



Figure 2 Existing Students' Car Park, Clifton Campus (Google Earth)

The existing car park is widely disliked because students and staff pay £2 a day to park. That revenue can be weighed against electricity value from the roof array. Using PVGIS, the 49 kWp south-facing array at 25° is estimated to generate about 46,924 kWh/yr at this latitude (European Commission Joint Research Centre, no date). After lighting and other non-IT equipment (about 6,400 kWh/yr in the design schedule), roughly 40,500 kWh/yr remains. Valued at the current non-domestic electricity price of 24.5 p/kWh (Department for Energy Security and Net Zero, 2026b), that is about **£9,900/yr** of avoided purchase (or a lower figure if exported under a feed-in tariff). By comparison, if about 120 spaces were used on average at 55% occupancy over 220 weekdays at £2/day, parking income would be about **£29,000/yr**. The PV alone does not fully replace that income, so the case for the project also rests on landscape, student facilities and encouraging sustainable travel—not only on selling power. A smaller, discounted or free car park could sit elsewhere on campus under solar canopies, while bus routes from campus to the city centre and the Rivergreen tram stop support arrival without a car.

Arrival at The Heat Exchange is designed around walk and cycle paths from the main road, with a secure cycle hub, showers and changing on the east end of the building.

Table 2 Site facts

Item	Existing	Proposed
Primary land use	Student / staff car park	Student learning hub and landscape
Approximate plot area	about 10,200 m^2 (proposal plot from the site model)	Same plot; building footprint about 400 m^2
Parking / cycle provision	Surface parking; some EV use with NTU ID	Cycle hub (40 spaces), e-bike charging; reduced parking elsewhere

Item	Existing	Proposed
Surface water	Hardstanding; limited soakaway	Soft landscape, pond / rain garden, semi-permeable paths

SECTION 2

Design concept

2.1 Overall concept and organisation

The Heat Exchange is a two-storey timber learning hub organised so that it fits into the Clifton Campus ecosystem. The concept is deliberately ambitious about transforming the surrounding landscape, while remaining practical for everyday users (Nottingham Trent University, no date). Taking inspiration from the Clifton Campus library, rooms are arranged by intended noise level: teaching and social spaces, focused study, and quieter wellbeing uses sit apart so people can choose the environment they need.

The main block is 24 m by 14.4 m on a 4.0 × 4.8 m structural grid, with a single-storey cycle hub (6 × 9.6 m) on the east. Public and daytime rooms face south toward campus and daylight; plant, servers and WCs sit in a north strip. A double-height atrium links the floors.

Table 3 Principal spaces

Code	Room	Floor	Area (m²)	Use
G01	Entrance atrium & café	Ground	57	Double-height entrance, café / reception
G04	Flexible teaching hall	Ground	91	About 48 seats; reconfigurable for events
G05	Makerspace	Ground	27	Prototyping, 3D printing, electronics
G09	Edge server room	Ground	17	IT load; waste heat to the heat pump
G11	Cycle hub	Ground	50	Secure two-tier parking and e-bike charging
F01	Breakout & co-working	First	40	Informal lounge around the atrium
F04	Quiet study library	First	91	Silent study seats and book stacks
F07–F10	Group study pods	First	8 each	Bookable four-person rooms with screens
G03	Wellbeing & prayer room	Ground	17	Quiet, bookable reflection / prayer
F11	Smart building control room	First	17	Live energy dashboard for teaching

2.2 Envelope, appearance and users

Externally the building reads as a timber construction with generous glazing and a **sawtooth roof** carrying solar panels on the south-facing slopes. The north faces of the teeth are glazed to bring soft daylight deep into the first floor without the glare of direct sun—the same logic used in many industrial north-light roofs, here combined with PV on the opposite slope.

Table 4 Envelope summary

Element	Proposal	Notes
Structure	Engineered timber (CLT) primary frame and floors	Stores carbon in the structure (Churkina et al., 2020)
External wall	about 360 mm overall	Timber / masonry outer leaf with hempcrete thermal wrap — see Section 4
Roof	Sawtooth; PV on 25° south slopes	North light + generation; 1.2 m south overhang shades summer sun

Element	Proposal	Notes
Glazing	Higher on south (about 49% WWR) and west; lower on north (about 20%)	Winter sun welcome on south; north kept for heat loss

Day-to-day use assumes regular classes, a student-run café open roughly 09:00–17:00 in term time, and accessible toilets. Inclusive design follows Approved Document M for non-domestic buildings: level entrance, 1.5 m corridors, a lift, and accessible WCs (HM Government, 2015). Fire strategy would follow Approved Document B (means of escape, compartmentation and detection appropriate to a teaching building) (HM Government, 2019). External doors would use motorised or “press to open” accessible operation consistent with NTU estates practice across campus; the exact ironmongery brand would follow the university’s approved supplier list. Non-gendered accessible WC provision supports Goal 5 (equality) in day-to-day use (United Nations, 2015).

SECTION 3

Sustainable technologies

3.1 On-site renewable generation

The sawtooth roof carries **114 photovoltaic modules**, **49 kWp** in total, on south-facing slopes at **25°**. For Clifton's latitude, PVGIS (PVGIS-SARAH3 climate data) estimates about **46,924 kWh/yr**, or roughly **958 kWh per kWp** (European Commission Joint Research Centre, no date). At 24.5 p/kWh (Department for Energy Security and Net Zero, 2026b) that generation is worth about **£11,500/yr** if used on site. An indicative installed cost of about £1,000 per kWp gives a one-off outlay of roughly **£49,000** for the roof array (excluding scaffolding and structural fixings), so simple payback against avoided electricity is on the order of four to five years before maintenance—indicative only, pending a contractor quote.

Grid connection of generation plant at this scale must follow the Distribution Network Operator process in Engineering Recommendation G99 (Energy Networks Association, 2023). The university would also need to clear the array through its estates and planning processes as a non-domestic rooftop installation. The sawtooth form is chosen so that PV sits on the sun-facing slope while the opposite face provides north light to the first floor—combining generation with daylight in one roof geometry.

3.2 Heat recovery from computing

Most buildings pay to cool computers and then pay again to heat rooms. Here a small **edge server room** (four racks at an assumed 3 kW IT load each) produces about **105 MWh/yr** of heat that would otherwise be rejected. Rear-door heat exchangers and a coolant loop feed a **water-source heat pump** in the plant room, raising the lukewarm water to about 45–55 °C for low-temperature heating, with a thermal store to buffer supply and demand. European energy-efficiency law increasingly expects waste heat from computing to be reused where feasible (European Parliament and Council of the European Union, 2023). The same idea is applied here at building scale. Heating demand still needs a full energy model before the heat pump can be sized finally; until then plant costs are marked "quote needed" in the cost schedule.

3.3 Lighting and controls

All teaching and study spaces use **LED panels** with **occupancy and daylight sensors**, so lights dim or switch off when rooms are empty or bright enough—matching common NTU campus practice. Against a fluorescent, always-on baseline, the design schedule uses about **4,021 kWh/yr** for lighting instead of about **18,923 kWh/yr**, a saving of roughly **79%** (about £3,650/yr at 24.5 p/kWh) (Department for Energy Security and Net Zero, 2026b). Wiring would follow standard low-voltage distribution with centralised lighting control (for example DALI-style dimming). Accessible door operators and sensor lighting are treated as part of everyday inclusive design, not as optional extras (HM Government, 2015).

SECTION 4

Materials, research and the SDGs

4.1 Materials strategy

Materials research for the team (summarised from the Sustainability Lead's presentation slides) compared several low-carbon options and then aligned them with how Clifton already looks.

Structure and outer leaf. Engineered **timber** remains the primary frame: it is strong, relatively light, and can cut embodied carbon substantially compared with steel and concrete (Churkina et al., 2020). For the visible outer leaf, the team also researched **K-Briq** units—unfired blocks made from near-100% construction debris, with manufacturer claims of about 30 N/mm² compressive strength, very low embodied carbon per unit, and an aesthetic close to conventional brick so the building can sit with existing campus fabric without painted cladding (Kenoteq, no date).

Walls and insulation. Hempcrete (hemp shiv with a lime binder) is proposed as the thermal wrap / infill: it combines insulation with moisture buffering, helps keep indoor humidity in a comfortable band, and is argued to be carbon-negative over its life as the crop grows and the lime carbonates (UK Hempcrete, no date). In combination with a timber frame and K-Briq outer leaf, the wall build-up targets the model's overall thickness of about **360 mm**.

Interior finishes. Recycled **carpet tiles** allow local replacement without full re-carpeting. **Demountable glass partitions** keep daylight moving through the plan and reduce waste if layouts change. **Zero-VOC paints** cut airborne chemicals at fit-out. Together these choices support healthier indoor air (Goal 3) and responsible materials use (Goal 12) (United Nations, 2015).

Table 5 Principal materials

Element	Material	Sustainability note	Source
Primary structure	Cross-laminated / engineered timber	Long-term carbon store vs steel/concrete	(Churkina et al., 2020)
Outer leaf	K-Briq (recycled aggregate block)	Unfired; high recycled content; campus-compatible look	(Kenoteq, no date)
Thermal wrap	Hempcrete	Insulation, humidity control, biogenic carbon	(UK Hempcrete, no date)
Flooring	Recycled carpet tiles	Replaceable modules; recycled content	Team research
Partitions	Demountable glass	Daylight; less demolition waste	Team research
Paint	Zero-VOC coating	Lower indoor emissions	Team research

4.2 Mapping to the Sustainable Development Goals

We address the Goals in three bands, matching the project aims (United Nations, 2015):

Table 6 Design features mapped to SDG targets

Band	Goals	Examples in this design	Targets (examples)
Health, education, equality	3, 4, 5	Daylight, IAQ-aware ventilation, wellbeing room; teaching / library / makerspace; accessible and non-gendered WCs	3.9, 4.7, 5.1

Band	Goals	Examples in this design	Targets (examples)
Resources and infrastructure	6, 7, 8, 9	Rain-soft landscape; roof PV and heat recovery; student jobs in the café; edge computing + smart control room	6.3, 7.2, 7.3, 9.4
Climate and partnership	13–17	Timber and low-carbon materials; landscape replacing hardstanding; open data and shared digital twin; campus / city transport links	13.3, 15.1, 17.17

The strongest technical claims sit with **Goals 7 and 9** (clean energy and innovative infrastructure) and **Goal 13** (climate action through fabric and landscape). Goals 4 and 5 are served by the learning programme and inclusive access rather than by plant alone.

SECTION 5

Cost analysis

5.1 Hypothetical budget and build-up

The brief asks teams to set a hypothetical budget so features stay consistent (Nottingham Trent University, no date). We adopted about **£2.4 million** for the whole project, built up as follows:

Table 7 Hypothetical capital budget

Package	Basis	Indicative £
Construction (shell and core)	720 m^2 GIA × about £2,800/m^2 (education-building order of magnitude from published cost services) (Royal Institution of Chartered Surveyors, no date)	2,016,000
Fit-out (priced catalogue items)	Itemised from the 3D room model	88,270
Roof PV array	about £1,000/kWp × 49 kWp	49,000
Plant allowance (heat pump, MVHR, battery, thermal store, rainwater)	Quotes still required; provisional sum	200,000
Working total		**about 2,353,000**

Fit-out is the part we can price in detail today. Every chair, desk, LED panel and sensor is counted from the model; unit rates are indicative supplier prices pending quotes. Furniture and IT dominate the priced total; major plant is still unpriced because its size depends on the heating model.

Table 8 Fit-out by category (priced items)

Category	Indicative cost (£)	Notes
Furniture and fittings	about 25,100	Includes café counter, study furniture
IT, AV and data	about 14,400	Excludes coolant unit (quote needed)
Making and teaching	about 12,600	Makerspace equipment
Cycling	about 12,700	Racks, lockers, e-bike charging
Sanitary and water	about 12,300	WCs, showers, refill
Lighting and controls	about 10,200	LED panels and sensors
Plant and energy (priced lines)	0	Heat pump, battery, MVHR await quotes
Total priced fit-out	**88,270**	About £123/m^2 GIA

5.2 Running costs and budget check

Electricity is costed at **24.5 p/kWh** (DESNZ non-domestic average including CCL, Q2 2026) and grid carbon at **0.131 $kgCO_2e$/kWh** (Department for Energy Security and Net Zero, 2026b; Department for Energy Security and Net Zero, 2026a).

Table 9 Annual electricity picture (modelled so far)

Flow	kWh/yr	£/yr at 24.5 p	tCO_2e/yr
Roof PV generation	46,924	11,496 (value if used on site)	6.1 avoided
Lighting (design)	4,021	985	0.5

Flow	kWh/yr	£/yr at 24.5 p	tCO_2e/yr
Other equipment (non-IT)	about 2,340	about 570	0.3
Server IT load	105,120	25,754	13.8
Space heating / ventilation fans	*to be modelled*	—	—

Arguments for the costs. The shell-and-core figure is large, but it buys a permanent learning building and a landscape upgrade on a plot the university already owns. PV payback against avoided grid electricity is comparatively short. Lighting savings of about £3,650/yr show that controls matter even before plant is finalised.

Arguments against / risks. Server electricity dominates running cost if it is counted fully against the building; net-zero operation is only plausible if that IT load is treated as a campus computing service whose heat is a by-product, and if heating demand is shown to match the recovered heat. Parking revenue of roughly £29,000/yr (see Section 1) is not fully replaced by PV value alone. Construction at £2,800/m^2 is a benchmark, not a tender. Until plant is quoted and an energy model exists, the £2.4 million envelope should be treated as a planning budget, not a fixed price.

SECTION 6

Method, evaluation and conclusion

6.1 How the project was developed

Once the team had chosen the Clifton car park, the lead engineer needed a way to put a real site into a 3D model. OpenStreetMap alone gives plan outlines but not terrain elevation (OpenStreetMap contributors, no date). After testing options, we used **Terra Printer**, which builds 3D terrain from open map and elevation sources for any place on Earth (Terra Printer, no date). From that export we flattened the car park, copied trees and greenery from the surroundings into the cleared area, and blocked out a building using forms taken from other campus buildings plus simple solids.

That gave a sketch of landscape and massing. For finer detail we moved the geometry into **Python**: rooms and structure as coordinates, then reusable objects (sofas, chairs, screens, solar panels) placed wherever needed. The printable files were produced through the **Bambu Studio** console API (Linux community build), iterating until the parts were sliceable. We printed the **site at 1:1000** and the **building at 1:100**. Combined print time is about **33 hours** and about **1.1 kg** of PLA across four colours (about £21 of filament at £20/kg).

As a team we were stronger at programming than at interactive CAD. Driving the model from code let us change dimensions once and regenerate prints, schedules and the web twin together. The trade-off is clear: placing objects by coordinates is slower than dragging in a GUI, and a small code error can corrupt an export—though the slicer usually warns before a bad print.

For the showcase poster we avoided a stack of paper boards. Each side of the 1:1000 site model has a slot for a **laser-engraved plywood wall** carrying key facts and the QR code to the web twin. That uses NTU workshop resources and folds the poster into the model itself. Overall, the method moved from a rough idea to printable site and building models within a few days of focused work.

Table 10 Contributions (edit names)

Section	Lead	Role
Section 1	*to be added*	…
Section 2	*to be added*	Design Lead
Section 3	*to be added*	…
Section 4	*to be added*	Sustainability & Research Lead
Section 5	*to be added*	Cost & Resource Coordinator
Physical / web models	*to be added*	Engineering / Visual Communications

6.2 Benefits and drawbacks

Benefits. Students gain a learning hub with quiet and social options; the campus gains green space aligned with its Green Flag landscape (Nottingham Trent University, 2026); energy systems demonstrate PV and heat reuse; and the project is explainable through physical models, a walkable web twin, Google Earth and engraved walls.

Drawbacks. Capital cost is high; parking income is not fully replaced by PV value; the west glazing needs careful summer shading; hempcrete and K-Briq detailing must be signed off by a structural engineer; and server electricity dwarfs roof generation if counted against the building alone.

Limitations. Heating demand and heat-pump size are not yet modelled; many plant prices are provisional; fit-out rates are indicative; and performance in use is unproven. Those limits bound any net-zero claim at the showcase.

6.3 Conclusion

The Heat Exchange proposes a two-storey timber learning hub on the Clifton Campus car park, with roof PV, waste-heat recovery, and a landscape that replaces hardstanding. It addresses SDG bands 3–5, 6–9 and 13–17 through education, infrastructure and climate-focused choices (United Nations, 2015), within a hypothetical budget of about £2.4 million. Next steps are a monthly energy model, supplier quotes for plant, and confirmation of materials detailing with estates. The plywood walls and models carry the quick story; this Design Outline is the referenced account required by the brief (Nottingham Trent University, no date).

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Appendix A

Supporting notes on revenue and PV assumptions

Table 11 Parking revenue estimate (illustrative)

Assumption	Value
Spaces represented by the existing car park (order of magnitude)	120
Weekdays per year	220
Average occupancy	55%
Fee	£2 / day
Estimated annual income	**about £29,000**

Table 12 PV surplus after non-IT building loads

Item	kWh/yr
PV generation (PVGIS, 49 kWp @ 25° south)	46,924
Lighting (design)	4,021
Other non-IT equipment	about 2,340
Surplus available to displace grid import	**about 40,500**
Value at 24.5 p/kWh	about £9,900

Server IT load (about 105 MWh/yr) is omitted from the “surplus” line because it is treated as a campus computing service in the heat-recovery concept; if it were charged fully to the building, there would be no annual surplus to export.